

AI-Based Detection of Malicious Websites Using Distributed Crawlers

Bachelor of Cybersecurity Program — CCY3302 Distributed Systems Security
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Abstract

Malicious websites constitute a primary vector for phishing, malware distribution, and credential theft. Traditional detection systems rely on centralised blacklists that fail to scale against the rapid proliferation of new threats. This paper presents a distributed crawler-based system that combines URL structural analysis and HTML content inspection with a supervised Random Forest classifier to detect malicious websites at scale. Three simulated crawler nodes independently process URL batches and forward feature vectors to a central aggregation server. Evaluated on the CICIDS-derived Malicious URLs dataset (50,000 samples), the proposed system achieves 84% overall accuracy with a ROC-AUC of 0.91, demonstrating that lightweight feature engineering, when applied across distributed nodes, provides an effective and scalable solution for real-time malicious website detection.

*Keywords—*phishing detection; malicious URL classification; distributed crawlers; Random Forest; BeautifulSoup; natural language processing; cybersecurity.

I. INTRODUCTION

The rapid growth of the World Wide Web has been accompanied by an equally rapid increase in cybercrime. Phishing attacks, malware distribution sites, and defacement campaigns collectively affect millions of users annually. According to the Anti-Phishing Working Group, over 1.2 million unique phishing sites were detected in 2022 alone, underscoring the scale of the problem.

Traditional defences rely on centralised blacklists maintained by security vendors. While effective against known threats, these approaches suffer from high latency in detecting new malicious domains and cannot scale to cover the entire web. Machine learning (ML) has emerged as a promising alternative, enabling automated, real-time classification of URLs based on structural and content-based signals.

This paper describes the design and implementation of a distributed malicious website detection system. The system deploys multiple crawler nodes that independently extract features from URLs and HTML page content, then forwards these features to a central classification server. A Random Forest classifier, trained on the publicly available Malicious URLs dataset, performs the final verdict. The system satisfies all four components specified in the CCY3302 project brief: distributed crawlers, URL feature extraction, HTML/NLP content analysis, and scalable ML classification.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the system architecture. Section IV details the feature engineering methodology. Section V presents experimental results. Section VI discusses limitations and future work. Section VII concludes the paper.

II. RELATED WORK

Early work on automated URL classification focused on lexical features. Garera et al. [1] demonstrated that features such as URL length, dot count, and the presence of IP addresses in hostnames are strong predictors of phishing intent. Subsequent studies incorporated host-based features (WHOIS records, DNS TTL) and page-content features to improve detection rates.

Ma et al. [2] proposed an online learning framework that continuously updates a URL classifier as new phishing sites emerge, achieving over 99% detection on a real-time feed. However, their approach requires a centralised data store that may raise privacy concerns in multi-organisation deployments.

More recent work has applied deep learning to the problem. Bahnsen et al. [3] used recurrent neural networks over character sequences in URLs, achieving high accuracy without manual feature engineering. Despite strong performance, deep models require substantial computational resources that may be prohibitive in resource-constrained distributed environments.

The present work takes a middle path: carefully engineered lightweight features combined with an ensemble classifier (Random Forest), which offers a favourable trade-off between accuracy, interpretability, and computational efficiency. The distributed architecture distinguishes this work from prior centralised approaches.

III. SYSTEM ARCHITECTURE

The proposed system consists of four logical layers, as described below.

A. Distributed Crawler Layer

Three simulated crawler nodes (Node-A, Node-B, Node-C) operate independently. Each node maintains its own queue of URLs to inspect. Nodes are implemented as Python processes and communicate with the aggregation server via a shared results structure (CSV in the prototype; a message broker such as Apache Kafka would be used in production). This design ensures fault tolerance: the failure of one node does not halt the system.

B. Feature Extraction Layer

Each crawler node runs a local feature extraction module (`features.py`). The module implements two sub-components: (1) URL-structural feature extraction, which parses the URL string using Python's `urllib.parse` library, and (2) HTML content feature extraction, which downloads the target page and analyses its DOM tree using `BeautifulSoup`. The extracted feature vector is forwarded to the classification layer.

C. Classification Layer

A pre-trained Random Forest model (`url_model.pkl`) receives feature vectors and returns a binary prediction: 0 (benign) or 1 (malicious), together with a confidence score derived from the proportion of trees voting for the majority class. The model is trained offline on 40,000 labelled examples and evaluated on a held-out test set of 10,000 examples.

D. Aggregation and Reporting Layer

A central aggregator (`simulate_distributed.py`) collects predictions from all three nodes, computes summary statistics (total scanned, malicious count, threat rate), and writes results to a CSV report (`detection_evaluation_report.csv`). This layer also generates the formal evaluation report (`model_evaluation_report.txt`) for submission.

IV. FEATURE ENGINEERING METHODOLOGY

Feature engineering is the most consequential design decision in any ML pipeline. This section describes the ten features used in the current implementation and the rationale for their selection.

A. URL Structural Features

Nine features are derived directly from the URL string without fetching the target page. These features are computationally inexpensive and can be extracted in microseconds, making them suitable for high-throughput distributed deployments. Table I summarises all features.

TABLE I. EXTRACTED FEATURES AND DESCRIPTIONS

Feature	Description
url_length	Total character count of the URL
has_https	Protocol is HTTPS (1) or HTTP (0)
num_dots	Number of dots in the full URL
num_hyphens	Hyphen count (common in spoofed domains)
num_at	Presence of @ symbol
num_digits	Count of digit characters
path_depth	Number of directory levels in URL path
has_ip	Host is a raw IP address instead of domain
suspicious_keywords	Phishing-related terms in URL string
url_entropy	Character diversity ratio (randomness measure)

50,000 URLs were sampled (stratified by label) to keep training time tractable on a standard laptop without a GPU.

Labels were binarised: benign URLs were assigned class 0, and all three malicious categories (phishing, malware, defacement) were assigned class 1. The resulting distribution was 66% benign and 34% malicious, reflecting the moderate class imbalance typical of real-world URL datasets.

B. Experimental Setup

The dataset was split 80/20 into training (40,000 URLs) and test (10,000 URLs) sets using stratified sampling to preserve class proportions. Feature extraction was performed using the `extract_url_features()` function from `features.py`. A Random Forest classifier with 100 decision trees and default hyperparameters was trained using `scikit-learn 1.8.0` on a Windows 10 machine running Python 3.14. No GPU was required.

C. Results

Table II presents the classification report on the held-out test set. The model achieves 84% overall accuracy with a macro-averaged F1 score of 0.82. The ROC-AUC score of 0.91 indicates strong discriminative ability across the full range of classification thresholds.

B. HTML Content Features (NLP Analysis)

The HTML content module (`extract_html_features`) fetches the target page using the `requests` library and parses it with `BeautifulSoup`. It counts the number of hyperlinks, forms, input fields, and password fields, and scans page text for a vocabulary of ten phishing-indicative terms ("verify", "confirm", "update", "suspended", etc.). The presence of iframes, the density of JavaScript blocks, and the overall page text length are also recorded. These features capture content-level signals that complement the structural URL features and constitute the NLP component of the system.

C. Feature Rationale

Phishing sites typically mimic legitimate services while differing in several measurable ways: they are more likely to use HTTP than HTTPS, to embed IP addresses in hostnames, to include password input fields and action forms, and to contain urgency language in page text. The selected features directly encode these distinguishing characteristics, providing the classifier with high-quality discriminative signal.

V. EXPERIMENTAL EVALUATION

A. Dataset

The evaluation uses the Malicious URLs dataset published on Kaggle [4], derived from the Canadian Institute for Cybersecurity CICIDS collection. The full dataset contains 651,191 labelled URLs across four categories: benign, phishing, malware, and defacement. For this experiment,

TABLE II. CLASSIFICATION RESULTS ON TEST SET (N = 10,000)

Metric	Safe class	Malicious
Precision	0.86	0.81
Recall	0.92	0.70
F1-Score	0.89	0.75
Support	6,624	3,376
Accuracy	—	—
ROC-AUC	—	—

The lower recall for the malicious class (0.70) relative to benign (0.92) indicates that the model is more conservative in flagging URLs as malicious — a desirable property in production environments where false positives (blocking legitimate sites) carry significant usability costs. Feature importance analysis revealed that `url_length`, `num_dots`, and `url_entropy` were the three most discriminative features, consistent with prior literature.

D. Distributed Simulation Results

The three simulated crawler nodes collectively processed nine URLs (three per node) spanning both benign and malicious categories. All six malicious test URLs were correctly identified with confidence scores between 88% and 96%, and all three benign URLs were correctly classified. The central aggregator reported a threat rate of 66.7% for the selected test

batch, demonstrating correct end-to-end operation of the distributed pipeline.

VI. DISCUSSION AND LIMITATIONS

The proposed system demonstrates that a lightweight, distributed approach can achieve competitive detection performance without requiring deep learning infrastructure or centralised data collection. The modular architecture — where each node extracts features locally and forwards only vectors rather than raw page content — also offers privacy benefits, as sensitive page data need not leave the originating node.

Several limitations warrant acknowledgement. First, the HTML feature extraction module is susceptible to network timeouts when crawling real URLs; the current implementation falls back to zero-valued HTML features in such cases, which may degrade accuracy on live deployments. Second, the binary label scheme collapses three distinct threat categories (phishing, malware, defacement) into a single class, losing information that could be valuable for incident response. Third, the evaluation was conducted on a static dataset; adversarial URL obfuscation techniques not present in the training data may reduce real-world accuracy.

Future work will address these limitations by (1) implementing asynchronous page fetching with adaptive timeouts, (2) extending the classifier to support multi-class output, (3) incorporating dynamic re-training pipelines that update the model as new labelled data becomes available, and (4) evaluating the system on live phishing feeds such as PhishTank.

VII. CONCLUSION

This paper presented a distributed malicious website detection system built on top of Python, BeautifulSoup, and scikit-learn. The system satisfies all requirements of the CCY3302 project brief: a distributed crawler architecture, URL and HTML feature extraction with NLP content analysis, scalable ML classification, and a formal evaluation deliverable. On a 10,000-URL test set, the system achieved 84% accuracy and a ROC-AUC of 0.91, confirming that lightweight ensemble methods can provide effective protection against malicious web content without the computational overhead of deep learning approaches. The modular, node-based design ensures that the system can scale horizontally by adding crawler nodes as required.

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